

In network fundamentals:

- Network
 - Switch
 - Routers
 - NAT
 - Internet Gateways
 - CIDR
 - Routing Table
- Etc.

AWS Network

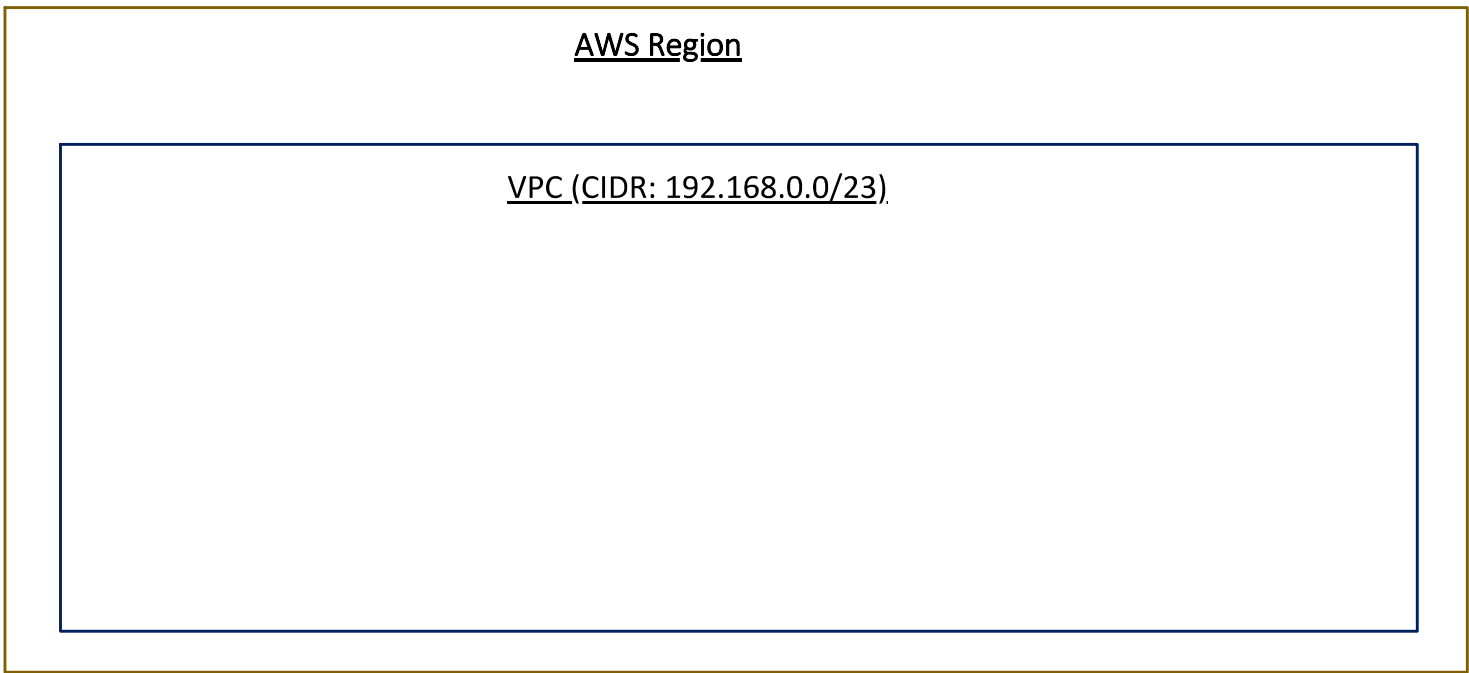
If we understood the fundamentals, then understanding AWS Network will be very easy:

In AWS terms, we have:

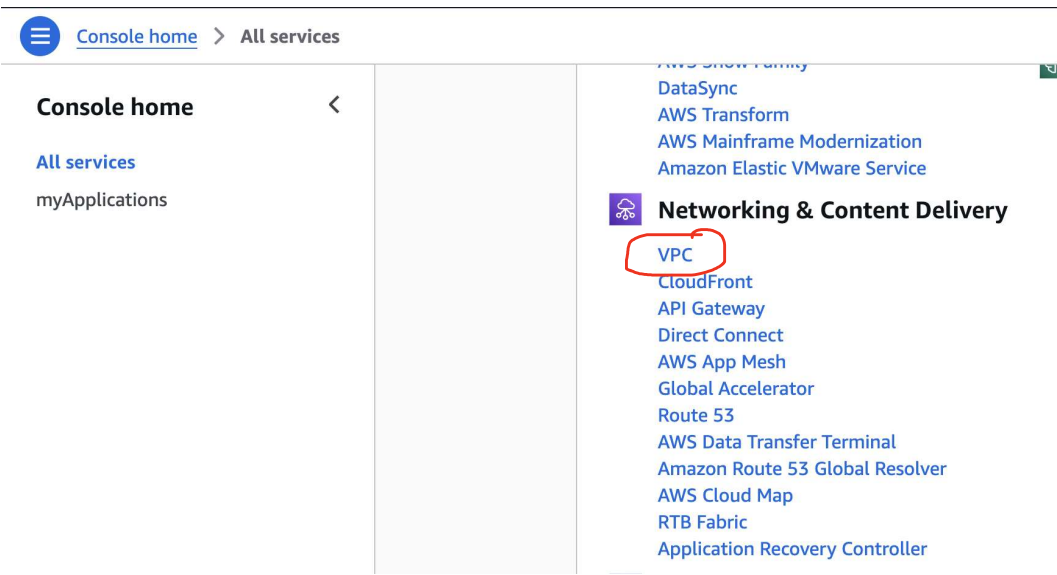
- VPC (Virtual Private Cloud)
- Subnet
- AWS Route tables
- NAT Gateways
- IGW (Internet Gateway)

VPC (Virtual Private Cloud)

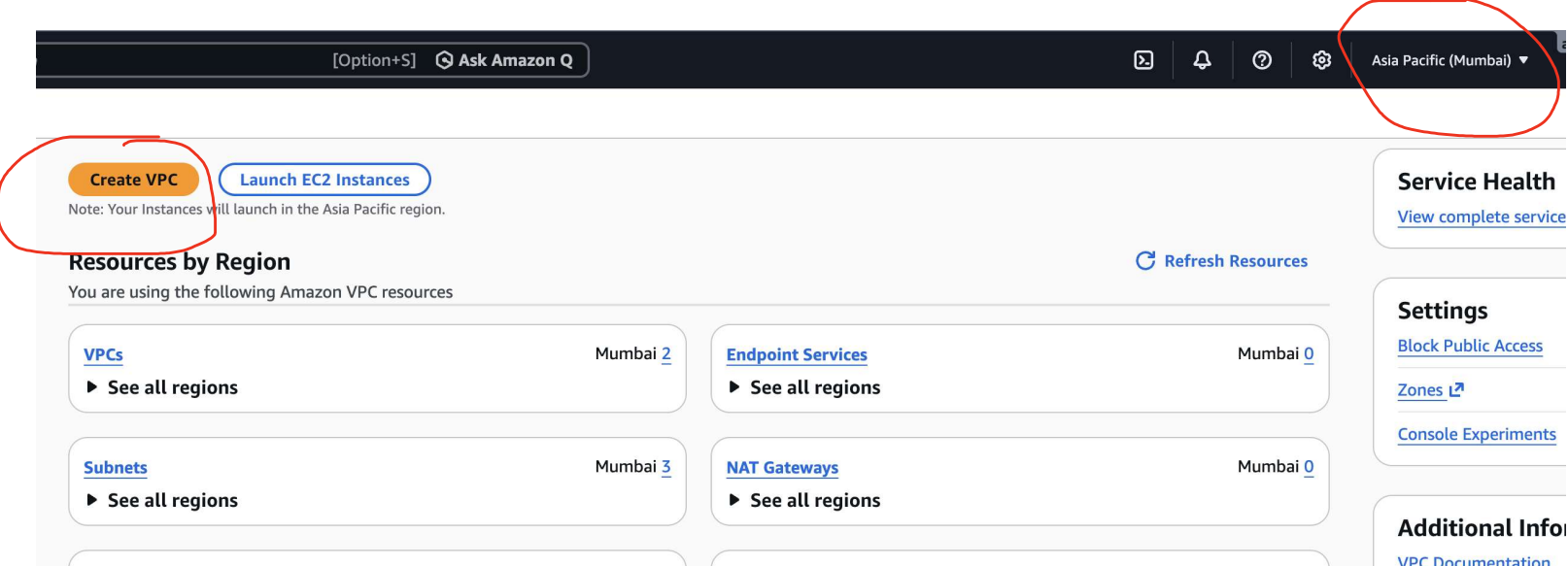
Its just a private network block defined by CIDR range.



All services -> VPC



Create VPC within a particular Region



Add details -> Create VPC

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional [Info](#)
Creates a tag with a key of 'Name' and a value that you specify.

vpc-conceptsbysravanish-prod

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
192.168.0.0/23
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

VPC encryption control (S) [Info](#)
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply US](#)

☒ None ☐ Monitor mode ☐ Enforce mode

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

[Remove tag](#)

You can add 49 more tags

Cancel

Preview code

Create VPC

VPC is created, also notice the Main (default) route table

vpc-0e288bbc758779d85

DNS resolution

Enabled

Main network ACL

acl-00683e4ac64f10b5d

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-00ec3ac7fa4633257

IPv4 CIDR

192.168.0.0/23

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-064c0c3c03f4370fe

IPv6 pool

-

Owner ID

643537035131

Routes (1)

Filter routes

192.168.0.0/23

local

Active

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

VPC

Your AWS virtual network

vpc-conceptsbyshrayansh-prod

Subnets (0)

Subnets within this VPC

Route tables (1)

Route network traffic to resources

rtb-064c0c3c03f4370fe

Network Connections (0)

Connections to other networks

Show all details

AWS Region

VPC (CIDR: 192.168.0.0/23)

Default Route Table

Destination	Target
192.168.0.0/23	local

Subnet

Big Network is divided into small sub networks.

For example:
VPC: 192.168.0.0/23
Means we can have . Means around 500+ hosts (some IPs are reserved for broadcast etc.) can be added in 1 VPC.

Now we can create small sub networks say around 100 IP address in each Subnet.

192.168.0.0/23 = 2^9 = 512

11000000 . 10101000 . 00000000 . 00000000

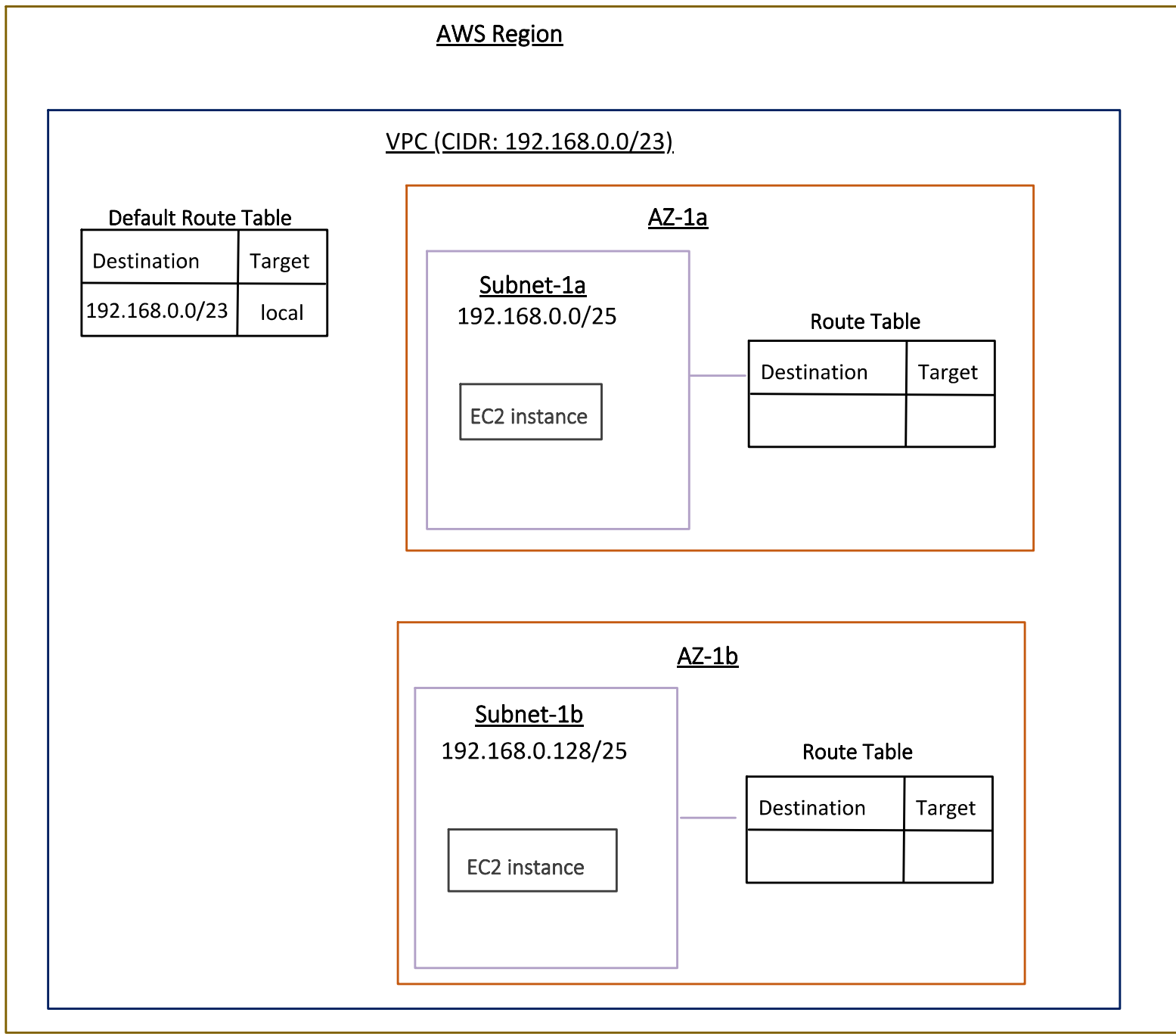
I want around 100 IP addresses in 1 subnet

2^6 = 64 (too small, we can not fit 100 hosts)
2^7 = 128 (perfect fit)

i.e /25 = 2^(32-25) = 2^7 = 128

Subnet	CIDR	IP Range	Total IPs	(fixed by VPC) (fixed by subnet) (free for hosts)
Subnet-1	192.168.0.0/25	192.168.0.0 to 192.168.0.127	128	11000000 . 10101000 . 00000000 . 00000000 11000000 . 10101000 . 00000000 . 01111111
Subnet-2	192.168.0.128/25	192.168.0.128 to 192.168.0.255	128	11000000 . 10101000 . 00000000 . 10000000 11000000 . 10101000 . 00000000 . 11111111
Subnet-3	192.168.1.0/25	192.168.1.0 to 192.168.1.127	128	11000000 . 10101000 . 00000000 . 10000000 11000000 . 10101000 . 00000000 . 10111111
Subnet-4	192.168.1.128/25	192.168.1.128 to 192.168.1.255	128	11000000 . 10101000 . 00000000 . 11000000 11000000 . 10101000 . 00000000 . 11111111

Generally for each AZ, we create its own Subnet.



2 Types of Subnet:

Private Subnet

- Private Subnet routing table do not have entry for Internet Gateway.
- Any hosts present within Private Subnet, uses NAT Gateway to talk to Internet.

Mostly in private subnets we keep:

- EC2 instances
- Database

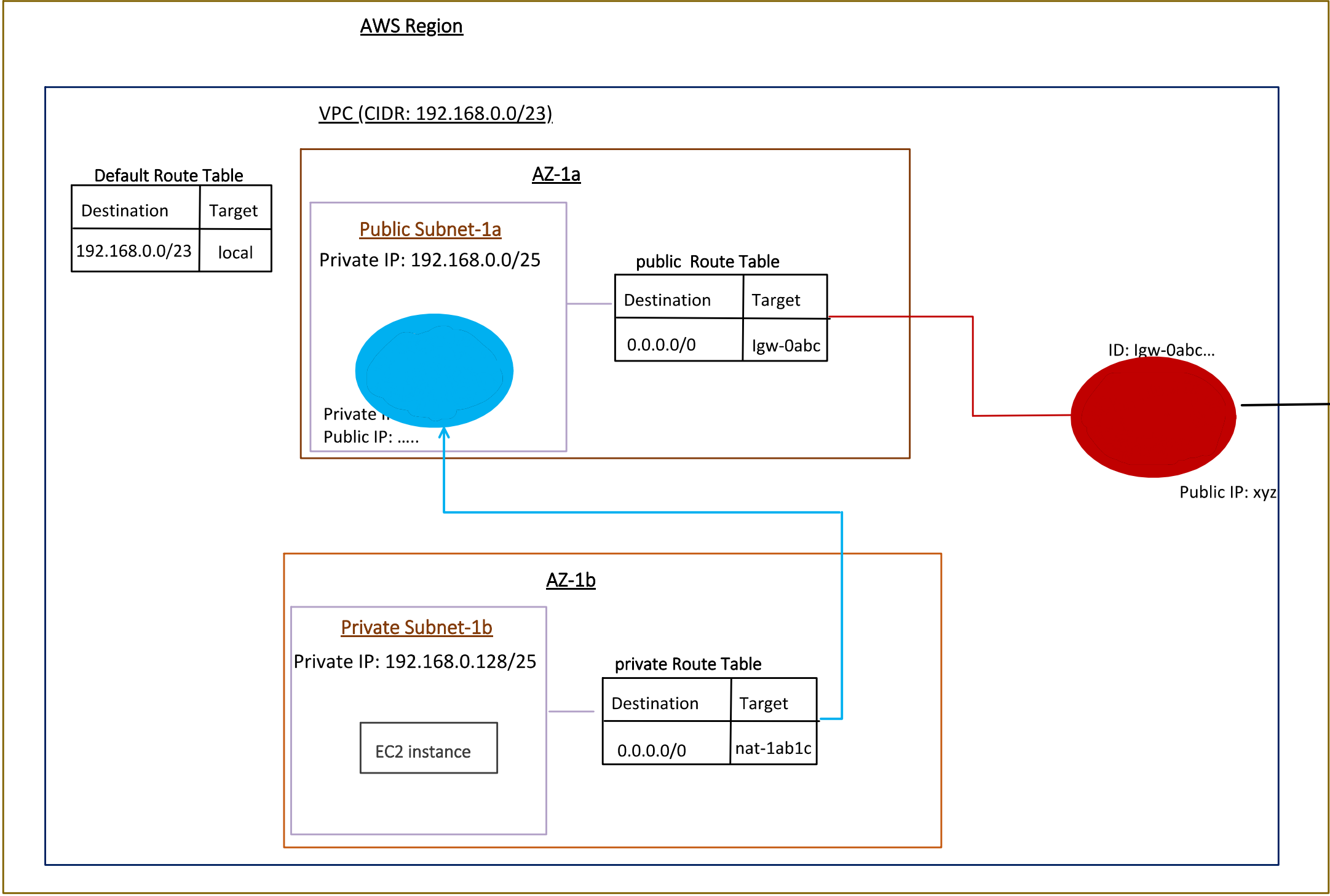
Public Subnet

- Public Subnet routing table do have entry for Internet Gateway.
- and any host (or EC2) sitting in this Subnet, also has public IP address and can talk directly with Internet.

Mostly in public subnet we keep:

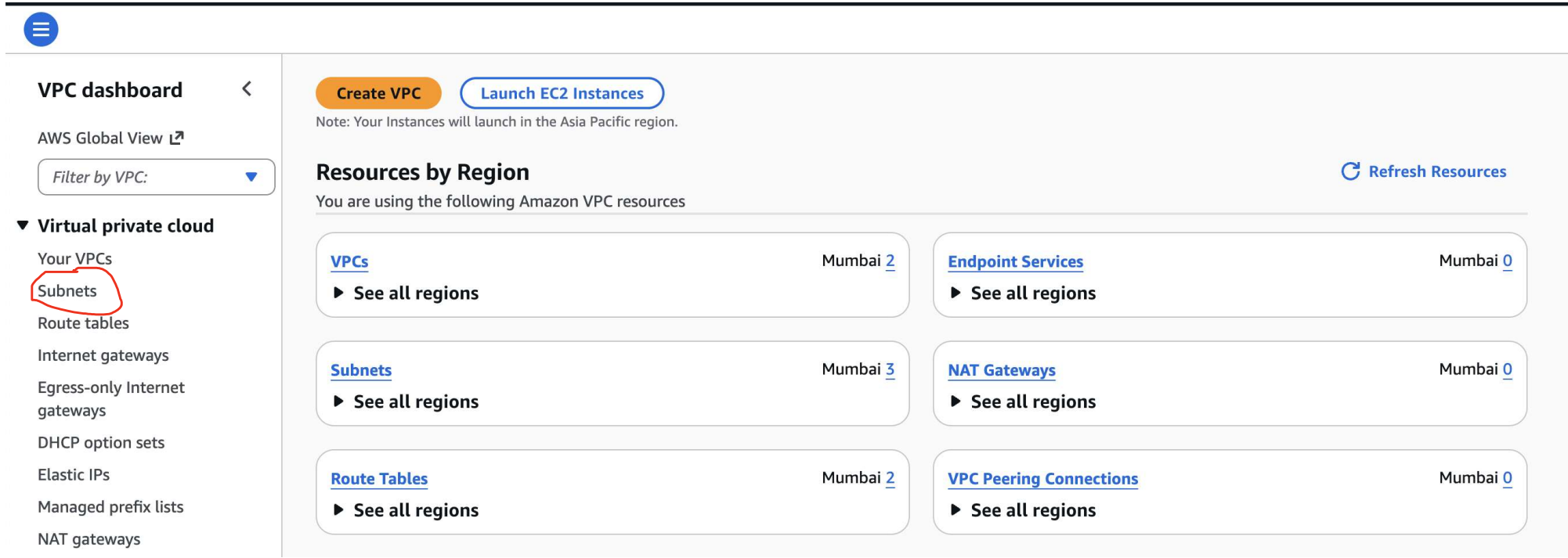
- Load balancer
- NAT Gateways

But there is not such hard rule, we can even create EC2 or database instance here.

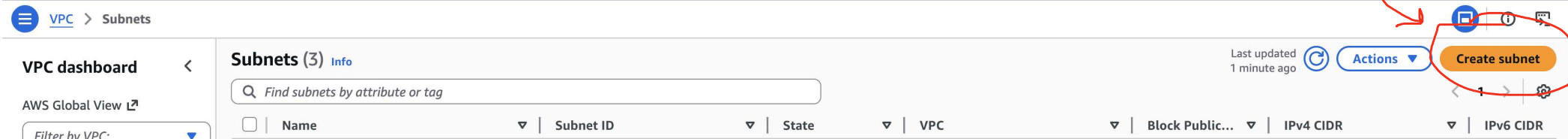


lets create subnet for AZ-1a

VPC -> Subnets



Click on Create Subnet



▼ Virtual private cloud						
Your VPCs	☐	-	subnet-0f3baa01f6bb5f1fe	Available	vpc-06a42df24a27c3d21	Off 172.31.32.0/20
Subnets	☐	-	subnet-09e56df6c66e24c67	Available	vpc-06a42df24a27c3d21	Off 172.31.16.0/20
Route tables	☐	-	subnet-075852bc286b9da6a	Available	vpc-06a42df24a27c3d21	Off 172.31.0.0/20
Internet gateways						
Egress-only Internet gateways						
DHCP option sets						
ELB subnets						

Select the VPC, in which this subnet need to be created

Create subnet

VPC

VPC ID

Create subnets in this VPC.

vpc-0e288bbc758779d85 (vpc-conceptsbyshtayansh-prod)

Associated VPC CIDRs

IPv4 CIDRs

192.168.0.0/23

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

public-subnet-1a

The name can be up to 256 characters long.

Availability Zone

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/23

IPv4 subnet CIDR block

192.168.0.0/25

128 IPs

▼ Tags - optional

Key

Q Name

Value - optional

Q public-subnet-1a

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

Since it's a public subnet -> any future EC2 instance created in this public subnet. Automatically public IPv4 will be assigned to it -> which EC2 instance will use it to talk to Internet Gateway

Select public subnet -> Actions -> Edit subnet settings

You have successfully created 1 subnet: subnet-06f64e033b2a4addc

Subnets (1/1)

Find subnets by attribute or tag

Subnet ID : subnet-06f64e033b2a4addc

Clear filters

checkbox

Name

Subnet ID

State

VPC

Block Public...

IPv4 CIDR

checkbox

public-subnet-1a

subnet-06f64e033b2a4addc

Available

vpc-0e288bbc758779d85 | vpc-...

Off

192.168.0.0/25

Actions

View details

Create flow log

Edit subnet settings

Edit IPv6 CIDRs

Edit network ACL association

Edit route table association

Edit CIDR reservations

Share subnet

Manage tags

Delete subnet

Click on "Enable auto-assign public IPv4 address" -> Save

Edit subnet settings

Subnet

Subnet ID

subnet-06f64e033b2a4addc

Name

public-subnet-1a

Auto-assign IP settings

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

checkbox

Enable auto-assign public IPv4 address

Enable auto-assign customer-owned IPv4 address

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

checkbox

Enable resource name DNS A record on launch

Enable resource name DNS AAAA record on launch

Hostname type

Resource name

IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

checkbox

Enable DNS64

Cancel

Save

Similarly, lets create private subnet in another AZ

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

private-subnet-1b

The name can be up to 256 characters long.

Availability Zone

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

192.168.0.0/23

IPv4 subnet CIDR block

192.168.0.128/25

128 IPs

▼ Tags - optional

Key

Q Name

Value - optional

Q private-subnet-1b

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

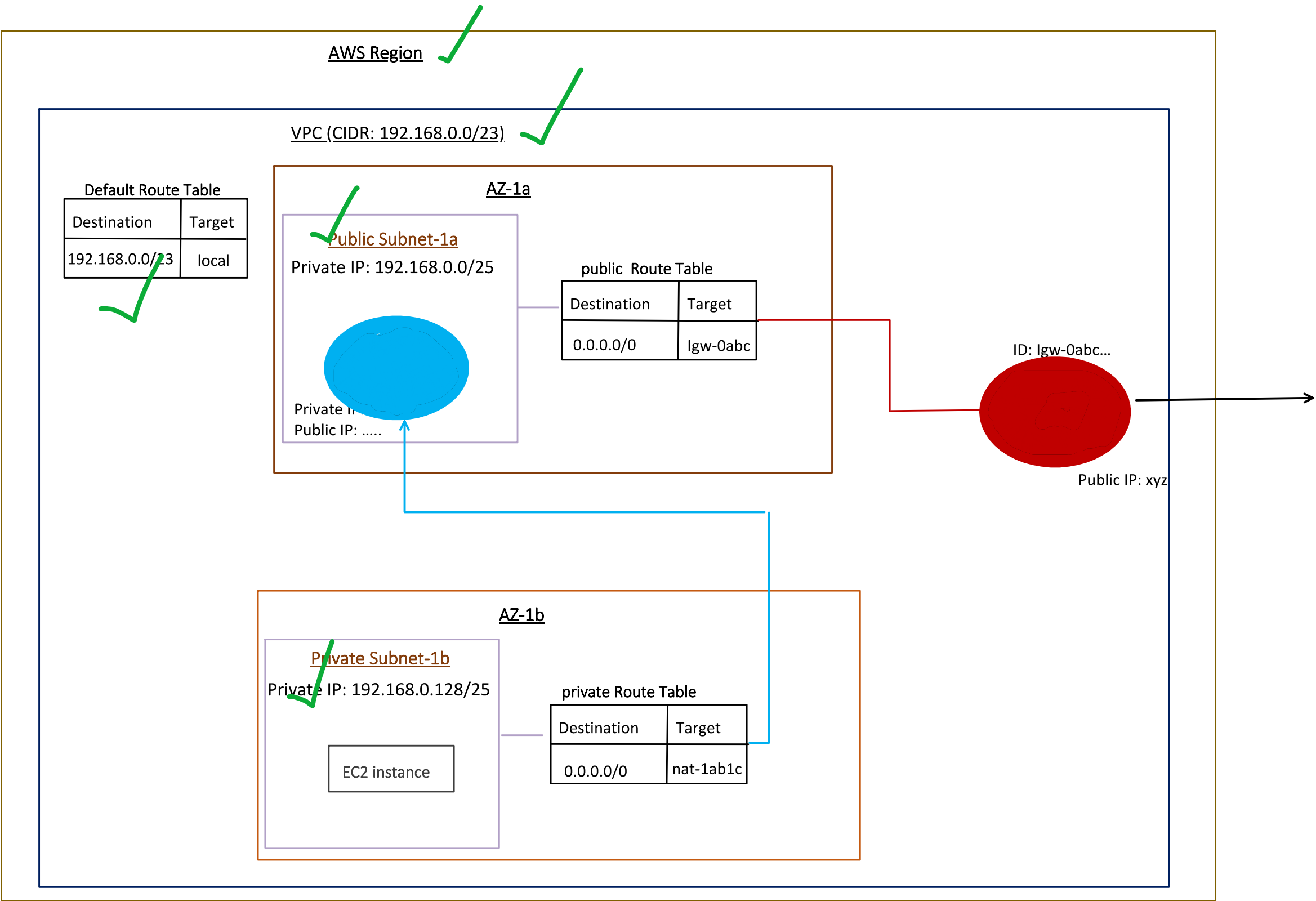
Create subnet

☐	public-subnet-1a	subnet-06f64e033b2a4addc	Available	vpc-0e288bbc758779d85 vpc-...	Off	192.168.0.0/25	-	-
☐	private-subnet-1b	subnet-0c36555e6d75397d8	Available	vpc-0e288bbc758779d85 vpc-...	Off	192.168.0.128/25	-	-

For our Private Subnet, we don't need Public IPv4 address for EC2 instances.

Pls Note:
Route table for each Subnet, we will set up later after set up of Internet Gateway and NAT Gateway.

Lets set up: INTERNET GATEWAY



Internet Gateway: it's a gate between our VPC and internet

- 2 things are required for anyone to talk to Internet Gateway:
- a. Subnet should have Internet Gateway route in its Route Table
 - b. Host should have Public IP

VPC -> Internet Gateways

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

Internet gateways

Create VPC

Launch EC2 Instances

Note: Your Instances will launch in the Asia Pacific region.

Resources by Region

Refresh Resources

VPCs

Endpoint Services

Subnets

NAT Gateways

Route Tables

VPC Peering Connections

Create Internet Gateway

Internet gateways (1)

Find Internet gateways by attribute or tag

Actions

Create internet gateway

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-0611fa34273f92d5f	Attached	vpc-06ae2df24a27c3d21	643537035131

Name -> create Internet Gateway

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag

Enter a tag with a key of 'Name' and a value that you specify.

conceptsbyshtayansh-internetgateway

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q conceptsbyshtayansh-internetgateway

Remove

Add new tag

You can add 49 more tags.

Cancel

Create internet gateway

Our newly created Internet Gateway -> Attach to VPC

The following internet gateway was created: igw-0f46d967b76d8d807 - conceptsbyshtayansh-internetgateway. You can now attach to a VPC to enable the VPC to communicate with the internet.

Attach to a VPC

igw-0f46d967b76d8d807 / conceptsbyshtayansh-internetgateway

Actions

Attach to VPC

Detach from VPC

Manage tags

Delete

Details

Info

Internet gateway ID

igw-0f46d967b76d8d807

State

Detached

VPC ID

-

Owner

645537035131

Tags (1)

Search tags

Key

Value

Name

conceptsbyshtayansh-internetgateway

Manage tags

< 1 > ⚙

Select our VPC and Click Attach

VPC > Internet gateways > Attach to VPC (igw-0f46d967b76d8d807)

Attach to VPC (igw-0f46d967b76d8d807)

Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

Q vpc-0e288bbc758779d85

AWS Command Line Interface command

Cancel

Attach internet gateway

Lets set up: NAT Gateway.

AWS Region

VPC (CIDR: 192.168.0.0/23)

Default Route Table

Destination	Target
192.168.0.0/23	local

AZ-1a

Public Subnet-1a

Private IP: 192.168.0.0/25

Public IP:

public Route Table

Destination	Target
0.0.0.0/0	lgw-0abc...

AZ-1b

Private Subnet-1b

Private IP: 192.168.0.128/25

EC2 instance

private Route Table

Destination	Target
0.0.0.0/0	nat-1ab1c

Internet Gateway

ID: lgw-0abc...

Public IP: xyz

NAT Gateway:

- It's a gate between Private Network and Public Network.
- Removes sender private IP (192.168.0.128) and replaces it with its own **Public IP**.
- Sends it out through the Internet Gateway to the web.
- When the response comes back, the NAT remembers who originally asked for it and forwards it back down the chain.

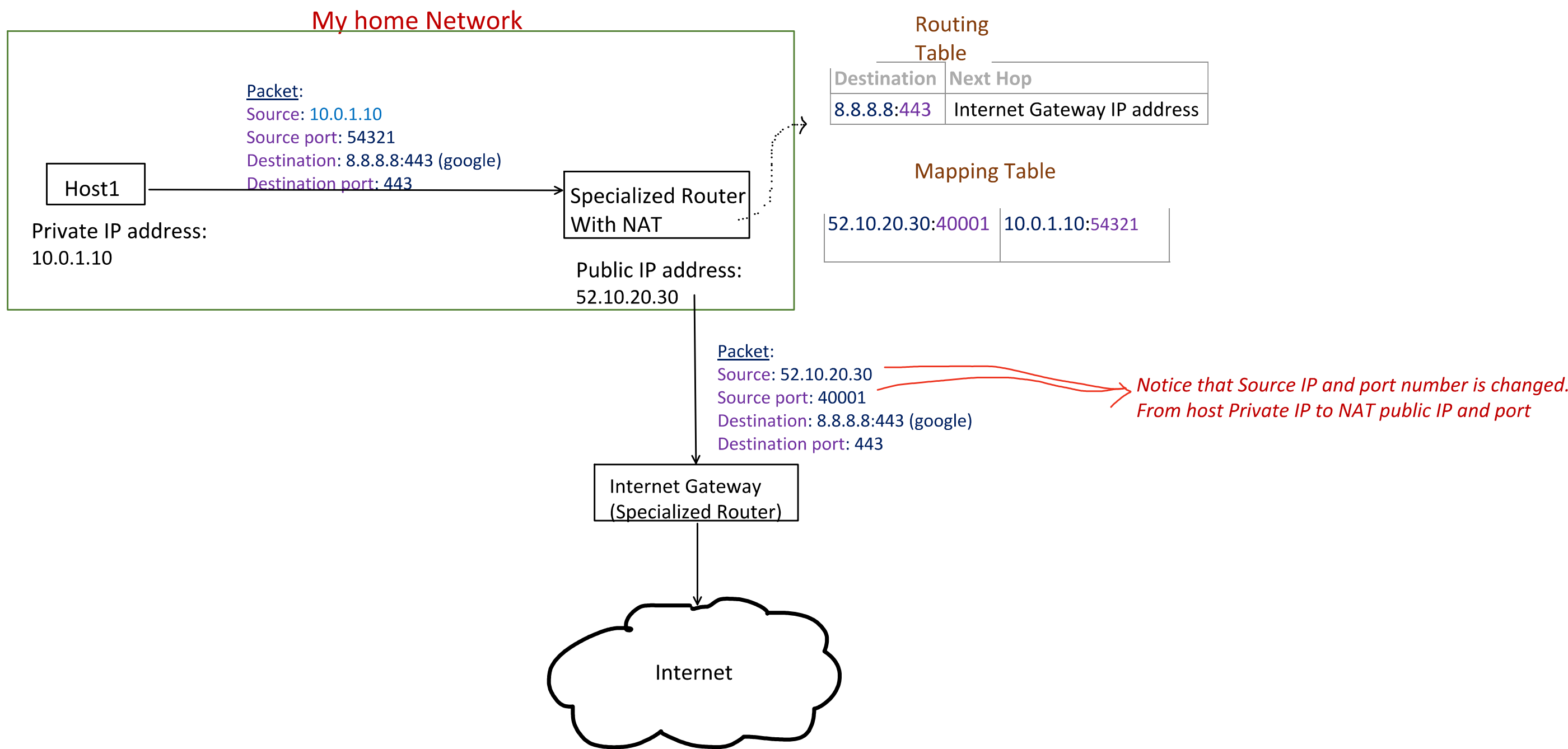
Another main Advantage of NAT is, its only support OUTGOING

Means, any packet coming from internet is DROPPED if there is no mapping present within NAT mapping table.

Or in other words

Talk must be first initiated by private network host only. Then only NAT will have entry in its Mapping table.

Remember this diagram from PART-1



VPC -> NAT Gateways

VPC dashboard

AWS Global View

Filter by VPC:

▼ Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways**

Create VPC **Launch EC2 Instances**

Note: Your Instances will launch in the Asia Pacific region.

Resources by Region

You are using the following Amazon VPC resources

VPCs Mumbai 2
► See all regions

Endpoint Services Mumbai 0
► See all regions

Subnets Mumbai 5
► See all regions

NAT Gateways Mumbai 0
► See all regions

Route Tables Mumbai 2
► See all regions

VPC Peering Connections Mumbai 0
► See all regions

Create NAT Gateway

NAT gateways info

Find NAT gateways by attribute or tag

Actions

Create NAT gateway

No NAT gateways found

Enter details:

NOTE: NAT needs to talk with Internet gateway that's why it should be present in Public Subnet only.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
conceptsbyshrayansh-NAT

Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.
☐ Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.
☒ Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.
subnet-06f64e033b2a4addc (public-subnet-1a)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID Info
Assign an Elastic IP address to the NAT gateway.
eipalloc-00005bb7c3633b262 **Allocate Elastic IP**

Additional settings Info

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
conceptsbyshrayansh-NAT

Allocate Elastic IP

Provide the name for NAT Gateway

Manually select the public subnet in which we want to put the NAT Gateway

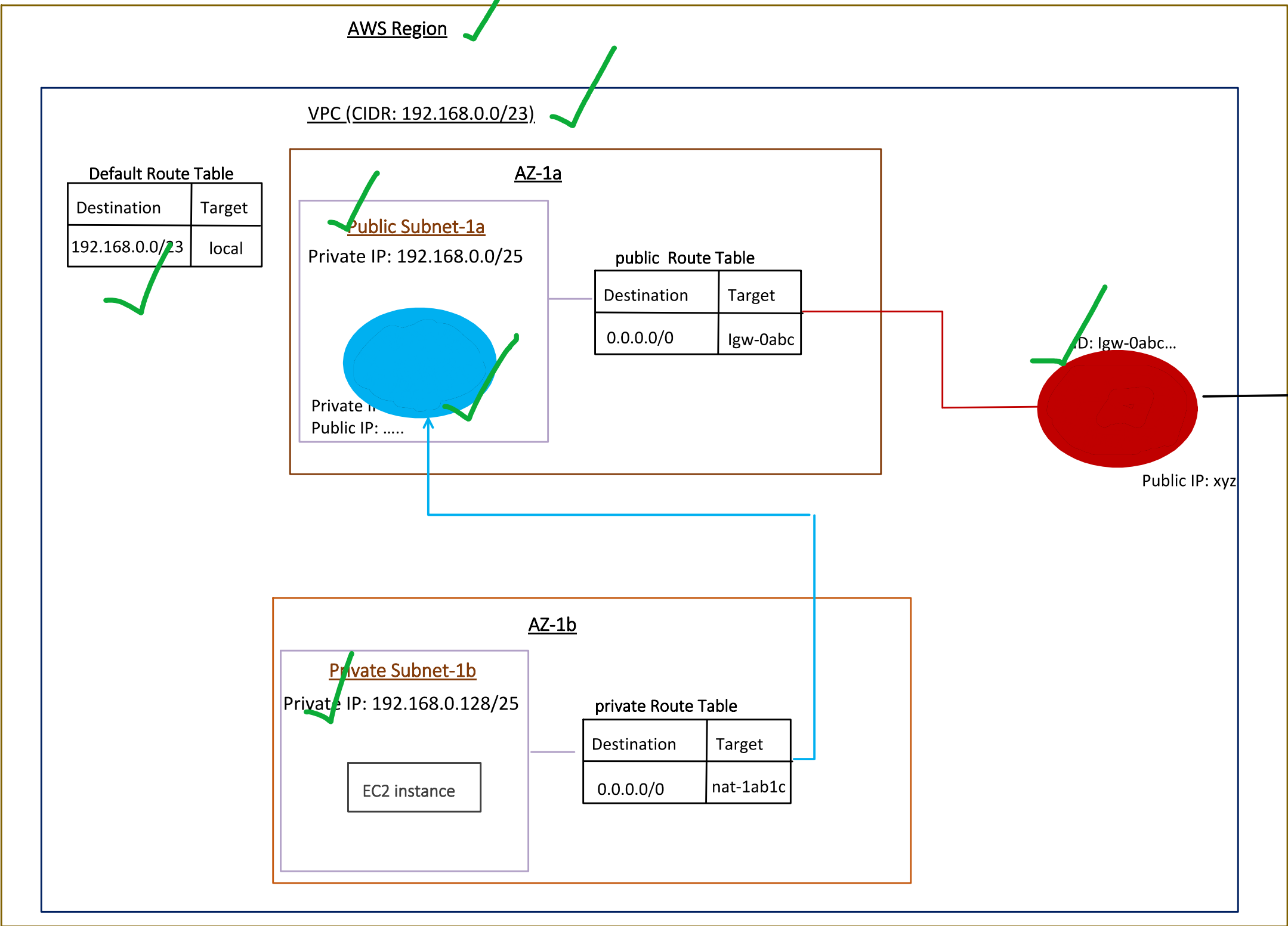
NAT Gateway need to talk to Internet Gateway That's why we need to put it inside PUBLIC SUBNET

- Allocate Elastic IP, this will provide stable public IP address. - it does not change even if service restarted.
- Why permanent bcoz, when a website replies it send back to NAT address, but if NAT public address changes frequently then its an issue.

Pls Note:

NAT Gateway and Elastic IP is both chargeable. So delete it if you don't intent to use it.

Lets configure Public Routing Table:



VPC -> Route tables

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Create VPC Launch EC2 Instances

Note: Your Instances will launch in the Asia Pacific region.

Resources by Region

You are using the following Amazon VPC resources

Refresh Resources

VPCs

See all regions

Endpoint Services

See all regions

Subnets

See all regions

NAT Gateways

See all regions

Route Tables

See all regions

VPC Peering Connections

See all regions

Create Route table

Route tables (1/2)

Find route tables by attribute or tag

Last updated 1 minute ago

Actions

Create route table

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☒		rtb-064c0c3c03f437dfe	-	-	Yes	vpc-0e288bbc758779d85 vpc-...	643537035131
☐		rtb-037435253c27c04e6	-	-	Yes	vpc-06e42df24a27c3d21	643537035131

Provide name and choose VPC -> Click create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

public-subnet-routetable

VPC

The VPC to use for this route table.

vpc-0e288bbc758779d85 (vpc-conceptsbyshtayansh-prod)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q public-subnet-routetable

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Since it's a public Route table -> need to add new entry in it for Internet Gateway

Route table rtb-031baf5458b24d0d4 / public-subnet-routetable

Details

Route table ID

rtb-031baf5458b24d0d4

VPC

vpc-0e288bbc758779d85 | vpc-conceptsbyshtayansh-prod

Main

No

Owner ID

643537035131

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Filter routes

Destination

Target

Status

Propagated

Route Origin

Both

Edit routes

192.168.0.0/23

local

Active

No

Create Route Table

VPCCRoute tablesrtb-031baf5458b24d0d4Edit routes

Edit routes

Destination192.168.0.0/23

Targetlocal

StatusActive

PropagatedNo

Route OriginCreateRouteTable

Add route

Cancel

Preview

Save changes

Add new route for Internet Gateway

VPCCRoute tablesrtb-031baf5458b24d0d4Edit routes

Edit routes

Destination192.168.0.0/23

Targetlocal

StatusActive

PropagatedNo

Route OriginCreateRouteTable

0.0.0.0/0

Internet Gateway

No

CreateRoute

Remove

Add route

Cancel

Preview

Save changes

Now we need to connect this Route Table with our Subnet

Action -> Edit subnet associations

Updated routes for rtb-031baf5458b24d0d4 / public-subnet-routetable successfully

Details

rtb-031baf5458b24d0d4 / public-subnet-routetable

Details

Route table IDrtb-031baf5458b24d0d4

VPCvpc-0e288bbc758779d85 | vpc-conceptsbysrjayansh-prod

MainNo

Owner ID643537035131

Explicit subnet associationssubnet-06f64e033b2a4addc / public-subnet-1a

Edge associations-

Actions

Set main route table

Edit subnet associations

Edit edge associations

Edit route propagation

Edit routes

Manage tags

Delete

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Both

Edit routes

1

DestinationTargetStatusPropagatedRoute Origin

0.0.0.0/0igw-0f46d967b76d8d807ActiveNoCreate Route

192.168.0.0/23localActiveNoCreate Route Table

Select with which Subnet we need to connect this Route table

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

public-subnet-1a

subnet-06f64e033b2a4addc

192.168.0.0/25

-

Main (rtb-064c)

private-subnet-1b

subnet-0c36555e6d75397d8

192.168.0.128/25

-

Main (rtb-064c)

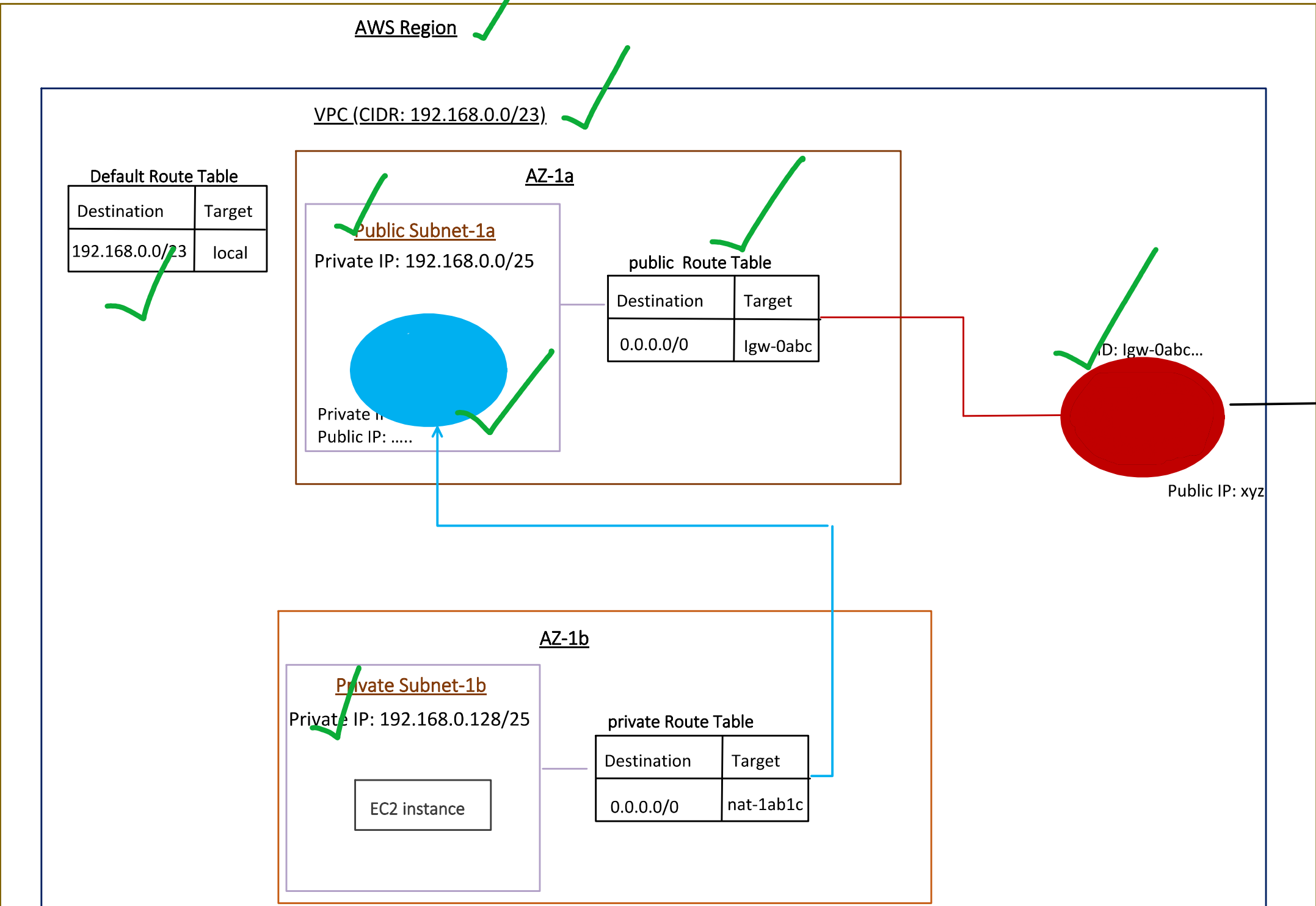
Selected subnets

subnet-06f64e033b2a4addc / public-subnet-1a

Cancel

Save associations

Public Route table is also set.



now only Private Route table we need to set.

Provide name and choose VPC -> Click create route table -> provide Name and Select VPC

Create route table

Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

private-subnet-routetable

VPC

The VPC to use for this route table.

vpc-0e288bbc758779d85 (vpc-conceptsbyshtayansh-prod)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q private-subnet-routetable

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Since this Route table is for Private Subnet, in this route table we need to add entry for NAT Gateway

Route table rtb-07ae65c12d63da233 | private-subnet-routetable was created successfully.

rtb-07ae65c12d63da233 / private-subnet-routetable

Details

Route table ID

rtb-07ae65c12d63da233

VPC

vpc-0e288bbc758779d85 | vpc-conceptsbyshtayansh-prod

Main

No

Owner ID

643537035131

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Filter routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/23	local	Active	No	Create Route Table

Both

Edit routes

VPC > Route tables > rtb-07ae65c12d63da233 > Edit routes

Edit routes

Destination

192.168.0.0/23

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

0.0.0.0/0

NAT Gateway

-

No

CreateRoute

Remove

Add route

Cancel

Preview

Save changes

Lets associate this Route table with our private Subnet.

Updated routes for rtb-07ae65c12d63da233 / private-subnet-routetable successfully

Details

rtb-07ae65c12d63da233 / private-subnet-routetable

Details

Route table ID

rtb-07ae65c12d63da233

VPC

vpc-0e288bbc758779d85 | vpc-conceptsbyshtayansh-prod

Main

No

Owner ID

643537035131

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0e210bf2529e460f8	Active	No	Create Route
192.168.0.0/23	local	Active	No	Create Route Table

Both

Edit routes

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	public-subnet-1a	subnet-06f64e033b2a4addc	192.168.0.0/25	-	rtb-031baf545i
☒	private-subnet-1b	subnet-0c36555e6d75397d8	192.168.0.128/25	-	Main (rtb-064c

Selected subnets

subnet-0c36555e6d75397d8 / private-subnet-1b

Cancel

Save associations

The diagram illustrates a multi-availability zone (AZ) VPC architecture designed for high availability. It is set within an **AWS Region**.

A **VPC (CIDR: 192.168.0.0/23)** is created, spanning two availability zones: **AZ-1a** and **AZ-1b**.

AZ-1a Configuration:

- Subnets:** Contains **Public Subnet-1a** (Private IP: 192.168.0.0/25) and **Private Subnet-1a** (Private IP: 192.168.0.128/25).
- Route Tables:**
 - Default Route Table:**

Destination	Target
192.168.0.0/23	local
 - public Route Table:**

Destination	Target
0.0.0.0/0	lgw-0abc...
- Resources:** A blue oval representing an EC2 instance is shown in Public Subnet-1a, with a green checkmark indicating it is healthy. A red oval representing an Internet Gateway (lgw-0abc...) is connected to the public Route Table.

AZ-1b Configuration:

- Subnets:** Contains **Private Subnet-1b** (Private IP: 192.168.0.128/25).
- Route Tables:**
 - private Route Table:**

Destination	Target
0.0.0.0/0	nat-1ab1c
- Resources:** A box representing an **EC2 instance** is shown in Private Subnet-1b, with a green checkmark indicating it is healthy.

Connectivity: A blue line connects the EC2 instance in AZ-1b to the EC2 instance in AZ-1a, ensuring traffic flow between the two availability zones.

VPC -> Select our VPC

vpc-0e288bbc758779d85 / vpc-conceptsbyshrayansh-prod
Actions ▾

Details info VPC ID vpc-0e288bbc758779d85 DNS resolution Enabled Main network ACL acl-00683e4c64f10b5d IPv6 CIDR (Network border group) — Encryption control ID —	State ✔ Available Tenancy default Default VPC No Network Address Usage metrics Disabled Encryption control mode —	Block Public Access Off DHCP option set dopt-00ec3ac7fa4633257 IPv4 CIDR 192.168.0.0/23 Route 53 Resolver DNS Firewall rule groups —	DNS hostnames Disabled Main route table rtb-064c0c3c03f4370fe IPv6 pool — Owner ID 643537035131
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[Resource map](#) | [CIDRs](#) | [Flow logs](#) | [Tags](#) | [Integrations](#)

Resource map [info](#)
⏻ Show all details

VPC

Your AWS virtual network

vpc-conceptsbyshrayansh-prod

Subnets (2)

Subnets within this VPC

ap-south-1a

A public-subnet-1a

192.168.0.0/25
No IPv6

ap-south-1b

B private-subnet-1b

192.168.0.0/25
Yes IPv6

Route tables (3)

Route network traffic to resources

rtb-064c0c3c03f4370fe

public-subnet-routetable

private-subnet-routetable

Network Connections (2)

Connections to other networks

conceptsbyshrayansh-internetgateway

conceptsbyshrayansh-NAT

```

graph LR
    AS[ap-south-1a public-subnet-1a] --- PR(public-subnet-routetable)
    PR --- IG(conceptsbyshrayansh-internetgateway)
    BS[ap-south-1b private-subnet-1b] --- PPR(private-subnet-routetable)
    PPR --- NAT(conceptsbyshrayansh-NAT)
  
```